

GenetiGraphCV: Computer Vision for Pedigree Charts

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Abstract—Pedigree charts remain a standard element of genetic documentation but are rarely utilized beyond record keeping due to the absence of automated digitization tools. This work presents a computer-vision based method for the recognition and basic interpretation of pedigree chart symbols. The proposed system detects male and female symbols using geometric analysis and determines affection status through greyscale intensity evaluation. A simplified symbol set is adopted to ensure consistent recognition under varying image conditions. The pipeline produces annotated visual outputs and tabular summaries suitable for integration with digital genetic data formats. Evaluation on representative samples confirms that deterministic geometric methods are sufficient for reliable symbol classification, though the system remains limited by fixed thresholds and the absence of relationship extraction. The approach provides a functional baseline for automated pedigree digitization and supports further development toward complete structural reconstruction and standardized data export.

I. INTRODUCTION

Pedigree charts are graphical representations that capture the genetic history of a proband with respect to a specific gene or disorder of interest. Although these charts have traditionally served as valuable tools in genetic analysis, they are increasingly being overshadowed by more advanced diagnostic techniques such as genetic screening and sequencing. As a result, pedigree charts are often collected as part of standard clinical protocols but are rarely utilized in diagnostics beyond a contributing factor. This has led to pedigree charts being relegated to records on file, without any widely available software for scanning and interpretation beyond simple digital photocopies.

There have been several developments in the field of genetic interpretations that require genetic data to be stored in structured digital formats. The authors in [2] proposed a model capable of interpretation and propagative analysis of genetic data, while also being robust to missing data. However, this model requires genetic information to be available in compatible digital formats rather than as image-based representations.

One such compatible representation is the PED file format [6], a structured tabular format—typically tab- or comma-separated—that records an individual’s affected status along with their familial relationships within the pedigree.

A key challenge in the digitization of pedigree charts arises from the wide variety of symbols and conventions

used across different representations. In this work, we adopt a simplification (subset) of standardized symbol set defined by the Pedigree Standardization Work Group of the National Society of Genetic Counselors, as discussed in Section III-A.

Despite the evident need for such a tool, no existing method currently facilitates the automatic conversion of pedigree charts into PED files. This paper explores the possibility of a simple computer-vision based solution for scanning and digitization of pedigree charts.

The remainder of this paper is organized as follows: Section II discusses related work, Section III details our digitization pipeline, and Section IV evaluates its performance.

II. LITERATURE REVIEW

A key part of the digitization of legacy medical records involves the conversion of physical data into digitally accessible formats, primarily as databases [4]. However, current Electronic Health Record (EHR) systems mostly focus on alphanumeric data due to its ease in digitization [7]. Other types of data such as medical image data and pedigree charts are preserved as-is, usually as digital scans.

The interpretation and analysis of medical image data is a heavily researched field that focuses on disease identification, data mining and pattern recognition. In contrast, pedigree charts have received negligible attention beyond simple scanning and storage as parts of patient histories. While this is symptomatic of the difficulty and uncertainty in the interpretation of genetic data, progress is steadily being made in the field of computational genetics that could benefit from a higher availability of data.

Some approaches to address this lack of data have focused on rebuilding of genetic histories through inferences made using other aspects of EHR data, such as parental lineages, birth history and obituaries [5]. Nevertheless, these approaches do not address the conversion and interpretation of pedigree charts themselves, which would significantly increase accessible and diagnostic confidence when genetic histories are of importance.

To the best of our knowledge, no software exists for the conversion of scanned pedigree charts into digital formats, namely PED files. In a paper published last year, Conte et al. discuss the possible architecture of such a system for

scanning pedigree charts [3]. They discuss several critical topics necessary for digitizing pedigree charts, involving symbol recognition, relationship inference and compilation of data for interpretation. Although comprehensive, the paper does not provide a functional implementation, leaving room for further research.

A. Related Problems

The challenge of automated pedigree chart digitization can be effectively decomposed into two fundamental, pre-existing computer vision problems:

- **Generic Symbol Recognition:** This involves the foundational task of accurately identifying the basic geometric shapes that constitute the pedigree chart, such as circles (females), squares (males), and diamonds (rotated squares denoting unknown sex). The complexity here lies in accounting for variations in drawing style, scale, and angle, which require robust classification algorithms.
- **Graph Digitization and Structure Extraction:** Beyond symbol recognition, the system must also process the relationships (lines, connectors, arrows) that link the identified symbols. This task is analogous to digitizing formal diagrams like Unified Modelling Language (UML) diagrams or flowcharts, where the primary objective is the extraction of meaningful, directional relationships and connectivity data between distinct elements.

III. METHOD

As explained in section II-A, the problems of scanning pedigree charts can be simplified into two sub-problems, out of which we shall be focusing on symbol recognition.

A. Simplifications

For a proof-of-concept system, we determined several simplifications that would serve to reduce the cardinality of the symbol set. These simplifications limit the realism of our system, but are an acceptable trade-off in the initial stages of development. Further research can focus on extending the symbol set, and adding multiple representations of symbols to account for deviations from conventional chart preparation guidelines.

We follow the conventions of Bennett et al. [1], but restrict our analysis to the fundamental symbol set: squares for males and circles for females, with filled shapes denoting affected individuals. Specialized markers (e.g., probands, carriers, or partial shading) are excluded to reduce variability and focus on geometric recognition. Figure 1 illustrates the simplified symbol set used throughout this study.

B. Symbol Identification

The first stage of our pipeline identifies occurrences of male and female symbols within the input image. Circles are detected using the Hough Circle Transform, which is robust to minor variations in boundary thickness and contrast. Squares, on the other hand, are identified using a custom contour-based algorithm (Algorithm 1) designed to detect quadrilateral regions that closely approximate true squares.

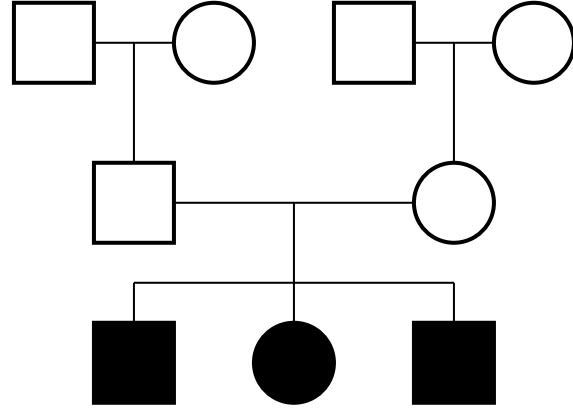


Fig. 1: Simplified example of a standard human pedigree chart following the conventions of Bennett et al. [1]. Squares represent males and circles represent females; filled symbols denote affected individuals, while unfilled symbols denote unaffected individuals. Other specialized symbols (e.g., proband markers or partial shading) are not considered in this work.

Algorithm 2 operates by first converting the image to greyscale and applying Gaussian blurring to reduce noise. Multiple threshold levels are then applied to capture both faint and strong edges. Each resulting binary mask is analysed for contours, which are polygonally approximated. Contours with four vertices, near-right angles, and approximately equal side lengths are retained as square candidates. A post-processing step filters duplicate detections based on centroid proximity, producing a final list of unique squares.

C. Rationale for Method Selection

The Hough Circle Transform was selected for circle detection because it is highly robust to variations in edge contrast and partial occlusions, which are common in scanned pedigree charts. Alternative methods, such as contour-based circularity measures or template matching, were considered but found to be more sensitive to noise and less consistent when symbol boundaries are imperfectly drawn.

For square detection, a custom contour-based approach was preferred over template matching or the Hough Line Transform because pedigree diagrams often include hand-drawn or rasterized symbols with slight geometric distortions. The contour-based method accommodates these irregularities while still enforcing geometric constraints (equal sides and near-right angles), offering a good balance between flexibility and accuracy. This choice also simplifies later integration with structural interpretation steps, where precise vertex locations are beneficial.

D. Symbol Analysis

After identifying all candidate symbols, we classify them as affected or unaffected based on their interior shading. Algorithm 2 describes this analysis procedure. For each detected symbol, the mean greyscale intensity within its region is computed; darker regions (intensity below a fixed threshold

Algorithm 1: Detection of Male Symbols (Squares)

Input: Image *img*
Output: List of unique square contours
Convert *img* to grayscale and apply Gaussian blur
Initialize empty list *all_squares*
for threshold *t* from 0 to 255 in steps of 26 **do**
 if *t* = 0 **then**
 Apply Canny edge detection and dilation
 else
 Apply binary thresholding with value *t*
 end
 Find all contours in the binary image
 foreach contour *c* **do**
 Approximate *c* by a polygon with tolerance $0.02 \times \text{perimeter}$
 if *c* has 4 vertices, sufficient area, and is convex **then**
 Compute bounding box (*w*, *h*) and aspect ratio $r = w/h$
 Compute maximum corner-angle cosine $\cos_{\max}$
 if $0.90 \leq r \leq 1.10$ and $\cos_{\max} < 0.15$ **then**
 Add *c* to *all_squares*
 end
 end
 end
end
Initialize empty lists *unique_squares* and *centroids*, distance threshold $d = 20$
foreach square *s* in *all_squares* **do**
 Compute centroid (x_s, y_s)
 if no stored centroid within distance *d* **then**
 Add *s* to *unique_squares* and store (x_s, y_s)
 end
end
return *unique_squares*

of 128) are interpreted as filled, corresponding to affected individuals.

Counts of affected and unaffected males and females are maintained throughout, and the results are visually annotated on the output image. Males are denoted by squares and females are denoted with circles, both coloured based on affection status, red for affected and green for unaffected.

IV. RESULTS

The implemented pipeline successfully detects and classifies individual symbols within standard pedigree diagrams. The system automatically distinguishes male and female symbols and determines their affection status based on interior shading. It further identifies generational groupings, enabling structural interpretation of the pedigree layout.

As illustrated in Figure 2, the output image is an annotated version of the input pedigree chart, in which each detected symbol is outlined and labelled according to its classification:

Algorithm 2: Pedigree Symbol Detection and Analysis

Input: Image *image*
Output: Counts of affected/unaffected males and females, annotated image
Load image and convert to grayscale
Initialize counters for male/female (affected, unaffected, total)
Set intensity threshold $T = 128$, generation gap $G = 30$
Detect male squares via *findMales*, female circles via *findFemales*
foreach male square *s* **do**
 Compute mean grayscale in *s*
 Mark as affected if mean $< T$, else unaffected
 Record centroid, bounding box, and update counts
end
foreach female circle *c* **do**
 Compute mean grayscale in *c*
 Mark as affected if mean $< T$, else unaffected
 Record centroid, radius, and update counts
end
Cluster individuals by vertical position (within G pixels) to assign generations
Sort individuals by generation and horizontal position
Label each individual sequentially and draw colored annotation:
red = affected, green = unaffected
return final counts, annotated image, and individual data list

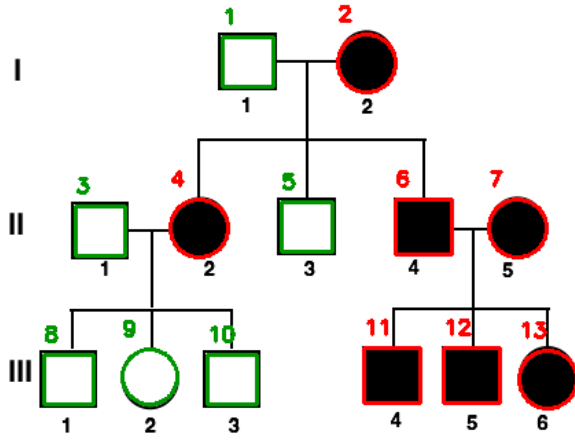
circles for females, squares for males, red shapes for affected individual and green shapes for unaffected individuals. This visualization provides a clear verification of the algorithm's performance and facilitates visual cross-checking against the original chart.

Evaluation on a small set of manually drawn and computer-generated pedigree charts indicates that the system achieves consistent detection performance under moderate variations in symbol size, line thickness, and image contrast. Occasional misclassifications are primarily associated with partially filled or degraded symbols, which remain challenging due to limited visual cues.

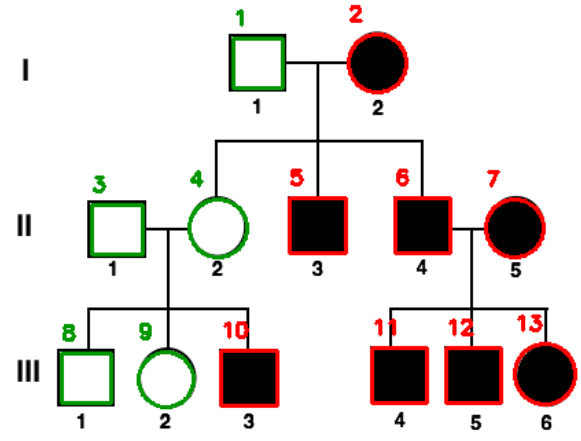
In addition to annotated visual output, the system produces a quantitative summary of the detected individuals, including total counts of affected and unaffected males and females, as well as their generation assignments. This information can be exported in a tab-separated values (TSV) file, providing a structured and portable representation suitable for downstream analysis or integration with external genetic data processing tools.

V. DISCUSSION

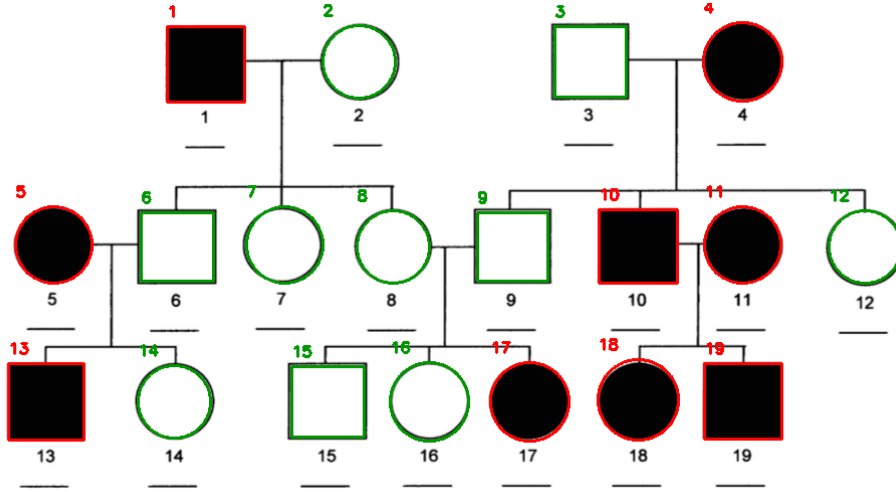
The results indicate that the proposed method performs reliable symbol detection and classification within the defined



(a) Sample 1 output.



(b) Sample 2 output.



(c) Sample 3 output (cropped).

Fig. 2: Annotated outputs from the analysis pipeline. Detected symbols are labelled according to sex and affection status: A.M. (affected male), U.M. (unaffected male), A.F. (affected female), and U.F. (unaffected female).

constraints. The combination of geometric and intensity-based analysis provides consistent performance across both scanned and synthetic pedigree charts. The Hough Circle Transform demonstrates stable detection for circular symbols even under moderate boundary irregularities, while the contour-based square detection algorithm maintains precision in identifying male symbols. The intensity thresholding approach successfully differentiates filled from unfilled symbols in most cases, producing stable affection classifications.

Several methodological limitations constrain the current system. The reliance on a fixed greyscale threshold (128) for determining affected status introduces sensitivity to illumination and contrast variations across input charts. Symbols with uneven shading or scanning artefacts may be incorrectly classified due to local brightness differences. This limitation suggests the need for adaptive or locally normalized thresholding

methods, which could dynamically adjust to regional intensity distributions and reduce misclassification under inconsistent lighting conditions.

The restricted symbol set further limits the method’s applicability. The exclusion of specialized symbols—such as probands, carriers, and partially shaded figures—reduces compatibility with clinical or archival charts that do not strictly follow standard drawing conventions. Additionally, the system does not yet infer relational links between symbols, preventing full reconstruction of family structures from detected entities. These constraints narrow the practical scope of the approach but establish a clear framework for future extension.

The present implementation demonstrates that deterministic computer-vision methods can perform stable recognition without learned models or extensive preprocessing. While simplified, this design supports reproducibility and serves as

a controlled baseline for later development of more adaptive, data-driven pipelines.

VI. CONCLUSION

This work demonstrates a functional approach to automated recognition of pedigree chart symbols using deterministic image processing. The pipeline detects and classifies male and female symbols and assigns affection status through intensity analysis. The method operates effectively within controlled conditions and yields structured outputs suitable for conversion into tabular formats.

Despite its limitations in symbol variety, illumination handling, and relationship extraction, the system establishes a reproducible foundation for digital pedigree interpretation. Future development should focus on expanding symbol coverage, implementing relationship detection for structural reconstruction, and introducing adaptive or learning-based thresholds for more robust shading analysis. These extensions would enable complete conversion of scanned pedigree charts into standardized digital representations such as PED files.

VII. AUTHOR CONTRIBUTIONS

P. N. Bhat and R. Sarin contributed equally to this work. P. N. Bhat was responsible for conceptualizing the study and experiments. R. Sarin performed the main programming and software development, with technical assistance from P. N. Bhat. Both authors equally contributed to the execution of the experiments and data acquisition. P. N. Bhat wrote the original draft manuscript. R. Sarin was responsible for reviewing, editing, and copy-editing the manuscript. Both authors read and approved the final manuscript.

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APPENDIX

AUTHORS' REPLIES TO REVIEWER COMMENTS

- 1) *A major strength of this report is its excellent literature review and clear problem framing. It identifies a practical research gap—the lack of automated tools to convert image-based pedigree charts into structured formats like PED files. The review is well-organized and effectively breaks the challenge into sub-problems such as “Generic Symbol Recognition” and “Graph Digitization,” which strongly supports the project’s phased methodology.*

Reply: The authors thank the reviewers for their positive evaluation.

- 2) *The report explains the methods used—such as the Hough Circle Transform for circles and a custom contour-based algorithm for squares—but it could be improved by providing more justification for these choices. Explaining why these specific techniques were preferred over other available options would make the methodology section more convincing and transparent.*

Reply: The revised manuscript includes an explicit rationale in Section III-C. This section now clarifies the selection of the Hough Circle Transform based on its robustness to boundary variation and partial occlusion, which are common in scanned or hand-drawn charts. The custom contour-based square detection method was chosen over alternatives such as template matching or

Hough-based line detection due to its tolerance of geometric distortion and its ability to preserve structural information for later interpretation. These clarifications strengthen methodological transparency and provide clear justification for each design decision.

- 3) *Including a discussion of methodological limitations would strengthen the report. For example, the system currently relies on a fixed greyscale threshold (≥ 128) to classify symbols as “affected,” which may be sensitive to lighting or contrast variations. Acknowledging such limitations and suggesting possible improvements (e.g., adaptive thresholding) would make the report more rigorous and forward-looking.*

Reply: Section V has been added to include a detailed treatment of methodological limitations. The discussion now explicitly acknowledges that the fixed greyscale threshold introduces sensitivity to lighting and contrast variability, potentially leading to inconsistent classification under uneven illumination or degraded scans. The revision further identifies adaptive thresholding and locally normalized intensity measures as promising approaches to mitigate these issues. In addition, broader constraints such as the reduced symbol set and lack of relational inference are discussed in relation to future work. These additions improve analytical rigour and situate the method as a validated baseline for subsequent technical refinement.